Experiment No : 3

Title: .Write a program in C++ to sort the numbers in an array using separate functions for read,display, sort and swap

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Program:

```
#include <iostream>
using namespace std;
void enterarray(int a[] , int n)
{
  cout << "Enter array a : ";
   for ( int i = 0 ; i < n ; i++)
   {
      cin >> a[i];
   }
}

void display(int a[] , int n)
{
   for ( int i = 0 ; i < n ; i++)
   {
      cout << "\n" << a[i] ;
   }
}

void sortascending(int a[] , int n)
{
   for ( int i = 0 ; i < n-1 ; i++)
   {
      for ( int j = 0 ; j < n-i-1 ; j++)
      {
         if ( a[j] > a[j+1])
          swap( a[j] , a[j+1]);
      }
   }
}
```

```
void sortdescending(int a[] , int n)
{
   for ( int i = 0 ; i < n-1 ; i++)
   {
      for ( int j = 0 ; j < n-i-1 ; j++)
      {
         if ( a[j] < a[j+1])
          swap( a[j] , a[j+1]);
      }
   }
}

void swap(int &x, int &y)
{
   int temp = x;
   x = y;
   y = temp;
}

int main()
{
   int a[100] , n;
   cout << "Enter number of elements (max 100) : ";
   cin >> n;
   enterarray(a , n);
   cout << "The original array is : ";
   display(a , n);
   sortascending(a , n);
   cout << "\nThe sorted array is (ascending) : ";
   display(a , n);
   sortdescending(a , n);
   cout << "\nThe sorted array is (descending) : ";
   display(a , n);

   return 0;
}
```

Output:

Output

```
Enter number of elements (max 100) : 6
Enter array a : 44 76 3 98 18 52
The original array is :  44 76 3 98 18 52
The sorted array is (ascending) :  3 18 44 52 76 98
The sorted array is (descending) :  98 76 52 44 18 3

=== Code Execution Successful ===
```